Alnaqbi and Ikuesan *Discover Computing* (2026) 29:202
https://doi.org/10.1007/s10791-026-10077-1

Discover Computing

REVIEW

Open Access

A systematic review of audio deepfake detection techniques for digital investigation

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Abstract
Deepfake technology has been driven by advanced machine learning and revolutionized multimedia creation by synthesizing hyper-realistic content. It includes images, videos, and audio. While its creative applications in entertainment and accessibility are significant, the technology also poses critical risks, especially in fraud, disinformation, and identity theft. Audio deepfakes are a subset of this phenomenon that replicate human voices with enhanced precision, mimicking tone, accent, and subtle vocal nuances. This has raised concerns in security-sensitive domains like voice authentication and forensic investigations. This systematic literature review (SLR) adopts PRISMA guidelines to explore the state-of-the-art in audio deepfake detection. It examines existing methodologies, features, datasets, and evaluation metrics across 27 studies. The review identifies traditional feature-based techniques like MFCC and LFCC, which, while effective, are limited by their dependency on manual engineering. In comparison, advanced frameworks, including CNNs, transformer-based architectures, and multimodal approaches, have achieved superior performance but often lack generalization in real-world scenarios. The findings highlight significant challenges, including dataset biases, adversarial vulnerabilities, and noise sensitivity, which limit scalability. Datasets such as FakeAVCeleb and DFDC achieved high accuracies but revealed performance inconsistencies due to dataset-specific characteristics. The review highlights the need for a generalizable audio deep fake detection framework that can achieve high accuracy across datasets.

Keywords Audio deepfake, Digital forensics, Machine learning, Multimodal analysis, PRISMA, Systematic literature review

1 Introduction

The extensive adoption of deepfake technology has introduced significant advancements as well as challenges within various sectors, from entertainment and social media to security and finance [1]. The availability of digital devices has increased the development of digital material, such as pictures and videos. The growth of social media has further increased the amount of multimedia content produced globally. Deepfakes were initially created to manipulate visual media, but with the growing technology, their use has been extended towards audio. In the audio fabrication, the deepfake uses sophisticated machine learning (ML) algorithms to accurately replicate human voices. In addition to just replicating the speaker's tone and accent, deepfake audio can also replicate subtle aspects of the speaker's voice, and that makes these artificial sounds nearly indistinguishable from authentic ones. Such capabilities present serious security implications in the fields that particularly require voice authentication. The potential applications for audio deepfakes are diverse, and they range from entertainment to accessibility, including automated voiceovers, virtual assistants, and personalized TTS (Text-to-Speech) systems. The ability of technology to mimic a person's voice with such precision also presents serious risks. Criminal applications of audio deepfakes, such as voice phishing, impersonation fraud, and disinformation, are on the rise [2]. A quite famous example of the threats posed by audio deepfakes occurred in 2019, when a deepfake audio clip was used to impersonate the CEO of a German parent company. In this case, the fraudulent voice commanded a UK-based energy company executive to transfer €220,000 to a fake supplier [3]. This incident highlighted the capability of synthetic audio to deceive individuals and how it has circumvented traditional security measures, resulting in severe financial consequences. Deepfake initially emerged in entertainment and have since taken over social media, where apps like FaceApp and Zao allow users to alter their appearances or insert themselves into popular movie scenes. Although this promotes creativity, it also raises ethical concerns [4]. As these tools become more advanced and accessible, it is now anticipated that similar cases of fraud and misinformation can occur. As a result, a surge has been observed in research focused on developing reliable, efficient, and scalable methods for detecting audio deepfakes. Most of them use machine learning and signal processing approaches. Audio media refers to any recorded or transmitted sound, such as speech, music, or sound effects, intended for human listening and typically stored in digital formats, including .wav, .mp3, or .aac.

1.1 Digital forensics: history, foundations and domains

Early progress in the field was focused on analyzing standalone systems to recover deleted files and examine activity logs. As the complexity of interconnected systems increased, it became crucial to use or devise specialized approaches to address cybercrime [5]. In 2001, the first Digital Forensic Research Workshop conference defined the field and created a roadmap for research and practice [6]. The workshop has highlighted the challenges that were posed by the evolving digital landscape, such as data complexity, hidden data recovery, and the trustworthiness of digital evidence. The events of September 11, 2001, have further demonstrated the urgent need for advanced DF capabilities. This has led to expanded research on securing digital infrastructures and addressing emerging threats like cyberterrorism and fraud [7]. It also gave rise to techniques such as hashing, cryptographic analysis, and forensic imaging, which allowed investigators to verify data integrity and authenticity [8]. By the early 2000s, mobile forensics emerged as a critical subfield and addressed the challenges posed by smartphones and other portable devices [9]. The science of DF extends to fields like multimedia forensics, where investigators analyze manipulated images, audio, or videos [10]. Techniques such as spectral analysis and ML help them to detect anomalies that are unnoticeable to the human eye. Similarly, in network forensics, packet analysis and intrusion detection systems are used to reconstruct events from network traffic [5]. There are several other subdomains that provide distinct patterns and processes in the analysis. There are several other subdomains that provide distinct patterns and processes in the analysis. Computer forensics is one of them, and it is the foundational subdomain of DF. It focuses on retrieving data from computers such as desktops, laptops, and external storage devices like hard drives or SSDs. Techniques such as data carving, metadata analysis, and file system examination are used to recover deleted files as well as to analyze logs and establish event timelines. Popular tools, such as EnCase [11] and FTK [12], are widely adopted in this domain to ensure data integrity during investigations. The subdomain of network forensics investigates network activities to detect malicious behavior, data breaches, and unauthorized access. In this field, analysis is performed on network packets, and communication events are reconstructed in a manner that makes it easy to identify anomalies. Tools like Wireshark [13] and NetWitness [14] are used by investigators to trace the origins of attacks and document findings for legal proceedings.

The rise of smartphones and portable devices has made mobile forensics an essential field. Several types of data, including call logs, messages, application data, and multimedia, have been extracted from devices in this domain. The challenges, such as high-level encryption and proprietary systems, are also taken into account during the process. Techniques such as chip-off analysis and logical acquisition are supported by tools like Cellebrite [15] and Magnet AXIOM [16], which are commonly used in investigations. Additionally, several challenges and complexities arise in distributed data storage within cloud environments. Therefore, the subdomain of cloud forensics usually focuses on the identification, preservation, and analysis of evidence in virtualized ecosystems. In this domain, the primary issues are typically multi-jurisdictional data and shared resource environments. Collaboration with cloud service providers and sometimes cross-border cooperation is acquired in this field [17]. Then there is the subdomain of multimedia forensics that specializes in analyzing digital images, audio, and video files to authenticate content and detect tampering. Techniques such as error level analysis, spectral analysis, and AI-based models are employed to detect manipulated media, including deepfakes. This subdomain is increasingly crucial in the modern misinformation scenario, as it verifies the authenticity of evidence and information [18]. Tools like X1 Social Discovery [19] play a significant role in overcoming these challenges.

Another most common and prevalent subdomain is Internet of Things (IoT) Forensics, which has now become critical due to the progression in smart technologies. IoT forensics examines data generated by interconnected devices, such as smart home systems, wearables, and industrial sensors. The most critical and severe challenges in this subdomain include handling diverse operating systems, fragmented data sources, and data volatility [20]. IoT forensics involves network-level, cloud-level, and device-level investigations to ensure the comprehensive collection of evidence.

While these remained the foundational areas of digital forensics, several other emerging subfields have significantly expanded the scope and sophistication of the forensic investigation domains. These subfields not only reflected the complexity of modern cybercrime but also addressed forensic challenges across domains that were previously overlooked. One of them is Behavioral Biometrics Forensics. It focuses on identifying individuals based on behavioral patterns, such as keystroke dynamics, mouse movements, and gait analysis. Unlike physiological biometrics, which rely on physical traits like fingerprints or iris patterns, behavioral biometrics offer continuous authentication and are less intrusive. As study [21] explained, these techniques are now increasingly used not just for access control but also in forensic investigations to attribute suspicious activity to specific users and support post-incident analysis. Forensic Readiness is another rapidly growing subdomain that has shifted the focus from reactive investigation to proactive preparedness. Traditionally, digital forensics takes place after an incident has occurred. However, forensic readiness encourages the preemptive configuration of systems to ensure that high-integrity data logs and audit trails are collected continuously and can be used as admissible evidence if needed. Study [22] proposed a forensic readiness framework that integrated behavioral biometrics into proactive forensic strategies, enabling systems to record user patterns in advance for later analysis during incidents. This subdomain has found relevance in sectors that require high accountability, such as finance and law enforcement. As discussed in ref. [23], forensic readiness ensures that digital systems are designed with evidentiary requirements in mind, thereby facilitating lawful interception, black-box auditing, and precise forensic recovery.

Another emerging area is Forensic Linguistics, which has been identified as a subfield of applied linguistics and can be studied as an intersection of language and law. It is particularly useful in analyzing written or spoken communication during criminal investigations [24]. This is usually helpful in incidents involving cybercrime, such as identifying the authorship of threatening emails, interpreting ambiguous legal documents, or verifying authenticity in online communication during fraud. As noted in ref. [25], this discipline has played a critical role in courtroom settings by offering insights into discourse analysis, legal language interpretation, and the presentation of linguistic evidence. Figure 1 shows the subdomains of DF.

Within these progressive forensic subdomains, one rapidly growing subdomain is audio deepfake forensics. The origins of audio deepfake technology lie in the progression of speech synthesis and voice cloning. TTS systems convert written text into speech and use neural networks (NN) to enhance the natural aspect and fluency. Prominent models, such as Tacotron 2 and Deep Voice, have refined the synthesis process by integrating linguistic and acoustic analysis modules to produce lifelike voice outputs [2].

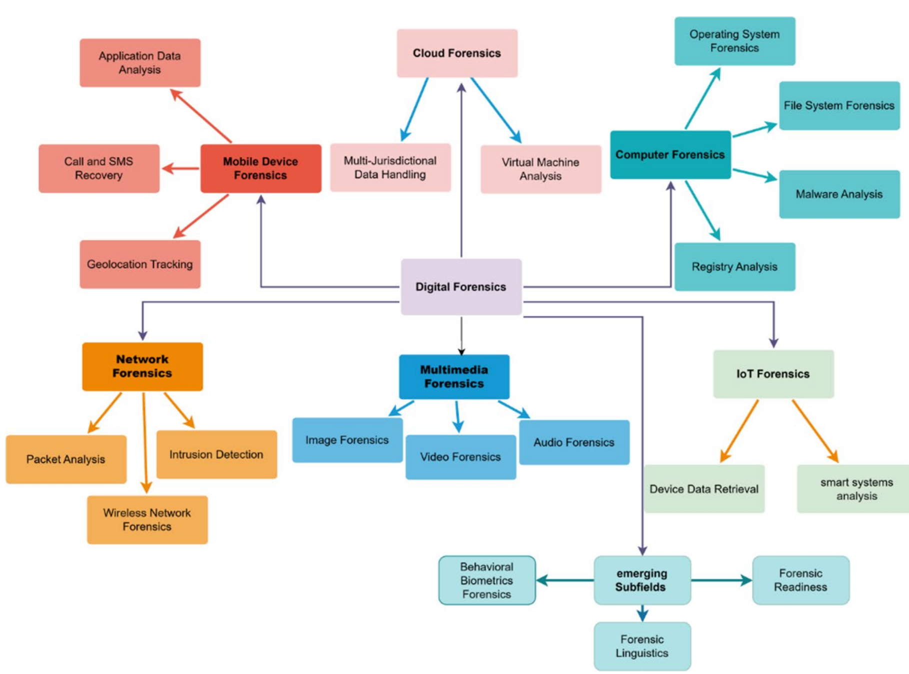

Fig. 1 Subdomains of DF research

Meanwhile, Voice Conversion (VC) frameworks focus on altering an individual's voice to sound like another person by manipulating features such as pitch, timbre, and prosody. These techniques often utilize parametric models and vocoders to reconstruct speech signals, maintaining a high degree of similarity to the target voice. The baseline of deepfake audio synthesis methods is rooted in Generative Adversarial Networks (GANs) and advanced speech synthesis models [26].

1.2 Existing reviews on deepfakes

The rapid expansion of deepfake generation technology has prompted a surge of systematic reviews aimed at mapping the fragmented detection landscape and establishing pathways toward robust countermeasures. Sharma et al. [27] developed a comprehensive taxonomy that spans image, video, and audio modalities by systematically categorizing algorithmic innovations, manipulation artifacts, and benchmark datasets, while documenting the persistent limitations in cross-domain generalization that continue to hinder practical deployment. This foundational work is substantively extended by Diwan et al. [28], whose meta-analysis of 260 video forgery detection studies provides rigorous comparative evaluation of deep learning versus traditional non-deep learning approaches, thereby illuminating advances in countering anti-forensic defenses—though the lack of standardized evaluation protocols significantly limits the consistency and reproducibility of cross-study benchmarking efforts. In the audio forensics domain, Bevinamarad et al. [29] systematically document the methodological transition from classical signal processing techniques to deep learning architectures, providing empirical evidence supporting hybrid models for enhanced detection fidelity; however, the review's 2020 publication date predates the emergence of diffusion-based audio synthesis methods that now represent a primary threat vector. Examining evidentiary applications, Al-Obaidi et al. [30] critically assess deepfakes' implications for legal and investigative processes, emphasizing the prevalence of CNN-based detection architectures while advocating for strengthened ethical governance frameworks, procedural standardization, and interdisciplinary collaboration—though this forensic orientation necessarily limits coverage of broader technological developments. Wu et al. [31] expand this scope by proposing a unified External/Internal taxonomic framework applicable to all AI-generated content modalities and introducing AI Safety Agents as proactive forensic mechanisms for preventive detection; however, the framework's theoretical foundation requires extensive empirical validation across diverse operational contexts to establish practical utility. Completing this review landscape, Rana et al. [32] provide systematic analysis of spatial, temporal, and frequency-domain features utilized in deepfake detection systems, presenting compelling evidence for multimodal fusion approaches to achieve operational robustness—yet the study's temporal scope, preceding the widespread adoption of diffusion models, positions it as an important historical reference rather than a comprehensive guide to addressing contemporary generative techniques.

1.2.1 Gap in existing review studies

Despite numerous systematic reviews addressing deepfake detection, critical methodological and conceptual gaps persist across modalities and research domains. Sharma et al. [28] develop a comprehensive taxonomy that spans image, video, and audio deepfakes, while documenting algorithmic approaches and benchmark datasets. However, their analysis exposes but fails to resolve the fundamental challenge of cross-domain generalization that impedes real-world deployment. Diwan et al. [28] extend this work through a meta-analysis of 260 video forgery studies, providing comparative insights into deep learning versus traditional methodologies. However, their findings underscore a more systemic deficiency—the absence of standardized evaluation protocols renders cross-study comparisons unreliable and prevents a meaningful assessment of detection advances. In the audio domain, Bevinamarad and Shirldonkar [29] systematically trace the evolution from signal-processing to deep learning approaches and advocate for hybrid architectures. However, their pre-2020 scope creates a critical temporal gap: the review predates the diffusion-based synthesis techniques that now dominate the threat landscape, leaving practitioners without guidance for contemporary challenges. From a disciplinary perspective, Al-Obaidi and Al-Obaidi [30] examine forensic and evidentiary implications, emphasizing CNN-based detection pipelines and calling for ethical frameworks and procedural standardization. However, their forensic focus creates a coverage gap that overlooks emerging detection paradigms from computer vision and multimedia security research. Wu and Wu [31] attempt cross-modal synthesis by proposing a unified External/Internal taxonomy for AI-generated content and introducing AI Safety Agents as preventive mechanisms; however, this work exemplifies a validation gap; the framework remains largely untested in operational settings, with limited empirical evidence regarding scalability or performance across diverse generation techniques. Similarly, Rana et al. [32] provide a foundational analysis of spatial, temporal, and frequency-domain features, while advocating multimodal fusion architectures; however, their pre-diffusion timeline limits their applicability to current state-of-the-art generators. Collectively, these reviews reveal interconnected deficiencies: generalization gaps between experimental and operational performance, standardization gaps preventing reproducible benchmarking, temporal gaps where reviews fail to address contemporary synthesis methods, scope gaps from disciplinary fragmentation, and validation gaps where theoretical frameworks lack empirical grounding—limitations that demand new approaches to cross-domain evaluation and accelerated knowledge transfer.

1.2.2 Contribution of the study

This study advances the field of audio deepfake detection by providing a systematic review and critical analysis of existing research methodologies, detection technologies, and benchmark datasets. Through a PRISMA-compliant structured approach, it addresses critical knowledge gaps by synthesizing empirical and methodological insights across the following dimensions:

- Comprehensive Survey of Contemporary Research: The review systematically examines state-of-the-art audio deepfake detection studies, identifying methodological trends, technological advancements, and emerging research directions within the field's evolving landscape.
- Comparative Evaluation of Detection Techniques: This work provides a rigorous comparative analysis of traditional machine learning, deep learning, and hybrid detection frameworks, systematically documenting the strengths, limitations, and operational constraints of each methodological paradigm for distinguishing synthetic from authentic audio.

- Feature Extraction and Indicator Analysis: The study conducts a comprehensive evaluation of acoustic features and forensic indicators employed in existing detection systems, providing evidence-based insights into which feature extraction methods demonstrate superior discriminative capacity for identifying manipulated audio content.
- Performance Benchmarking Across Datasets: This work evaluates detection model efficacy across diverse benchmark datasets through a systematic review of performance metrics, detection accuracy rates, and cross-dataset generalization capabilities, thereby identifying factors influencing operational robustness.
- Dataset Compilation and Quality Assessment: The study compiles and critically analyzes benchmark datasets commonly employed for audio deepfake detection research, evaluating their acoustic diversity, demographic representativeness, synthesis method coverage, and suitability for training generalizable detection systems.

The remainder of the paper is organized into 5 Sects. (2 to 6). Section 2 presents the methodology used to develop the systematic Literature review (SLR) of the study. Section 3 presents the literature review of the study in digital forensics and audio deepfakes. Section 4 presents the results and analysis of the study's findings, while Sects. 5 and 6 provide a comprehensive discussion and open research questions for future studies, respectively. Section 7 presents the conclusion and some recommendations for future work.

2 Methodology

The study has followed PRISMA [33] guidelines to conduct a systematic literature review on audio deepfakes. The guidelines ensure a structured method for devising a comprehensive search strategy development process, study selection, data extraction, and thematic identification. The systematic steps ensure a rigorous and reproducible review. The guidelines include several steps, as discussed in the section.

2.1 Research questions

This systematic literature review focuses on audio deepfake detection, aiming to synthesize current research and identify gaps in the field. The following research questions guide the investigation:

- RQ1: What are the existing techniques and algorithms discussed on audio deepfake detection?
- RQ2: What are the current features used in audio deepfake studies?
- RQ3: What are the audio deepfake indicators discussed in existing studies?
- RQ4: What are the current datasets and detection rate in existing studies?
- RQ5: What metrics are used to evaluate audio deepfake detection?
- RQ6: What are the challenges and limitations discussed in studies on audio deepfake detection?

2.1.1 Implementing search strategy

To capture a comprehensive set of studies on audio deepfake detection, a well-defined search strategy was implemented. The search aimed to identify all relevant literature focusing on detection techniques, datasets, features, performance metrics, and evaluation methods. The search was conducted across multiple academic repositories, including IEEE Xplore, ACM Digital Library, Springer, MDPI, and other reputable sources. The search terms were formulated using Boolean operators to ensure inclusivity and precision. The primary keywords and phrases included as: ("audio deepfake detection" OR "audio forgery detection" OR "deepfake forensics" OR "fake audio detection") AND ("existing studies" OR "literature review" OR "research") AND ("techniques" OR "methods" OR "algorithms" OR "approaches" OR "tools") AND (audio features" OR "acoustic features" AND ("detection rate" OR "accuracy" OR "performance" OR "evaluation" OR "success rate") AND ("metrics" OR "evaluation metrics" OR "performance metrics") AND ("datasets" OR "data sets" OR "benchmark datasets" OR "audio datasets") The search string was applied across the repositories, and filtration was performed based on the matching keywords, abstracts, and titles of studies.

2.1.2 Inclusion and exclusion criteria

Following the PRISMA guidelines [33], inclusion and exclusion criteria have been developed to include the relevant research material in the review, as shown in Table 1, while selected studies with references to the year are shown in Fig. 2.

2.2 Data extraction and study selection

The data extraction and study selection process followed systematic steps as discussed below:

Identification: An initial search retrieved 180 records across databases. Duplicate and irrelevant records (n = 50) were removed during the preliminary screening, leaving 130 records for further analysis.

Screening: The remaining records were screened based on relevance and availability. Nine studies were excluded due to retrieval issues, leaving 121 studies for eligibility assessment.

Eligibility Assessment: The studies were thoroughly reviewed based on predefined inclusion and exclusion criteria. After a detailed evaluation, 27 studies were deemed eligible and selected for inclusion in the systematic literature review. This rigorous selection process ensured that the studies included were directly aligned with the review objectives.

Data Extraction from Studies: The data extraction process involved systematically collecting relevant information from the selected studies to address the research

Table 1 Inclusion and exclusion criteria

Category	Criteria details
Inclusion criteria	Studies focusing on audio deepfake detection methodologies, techniques, and tools. Research reporting empirical results with measurable performance metrics such as accuracy, detection rates. Articles discussing the use of datasets or features specific to audio deepfake detection. Studies that are in the English language for inclusivity. Studies published between 2018 and 2024.
Exclusion criteria	Irrelevant abstracts or book chapters. Studies exclusively focused on video or image deepfakes. Review articles, theses, or conceptual frameworks without empirical findings. Articles in languages other than English.

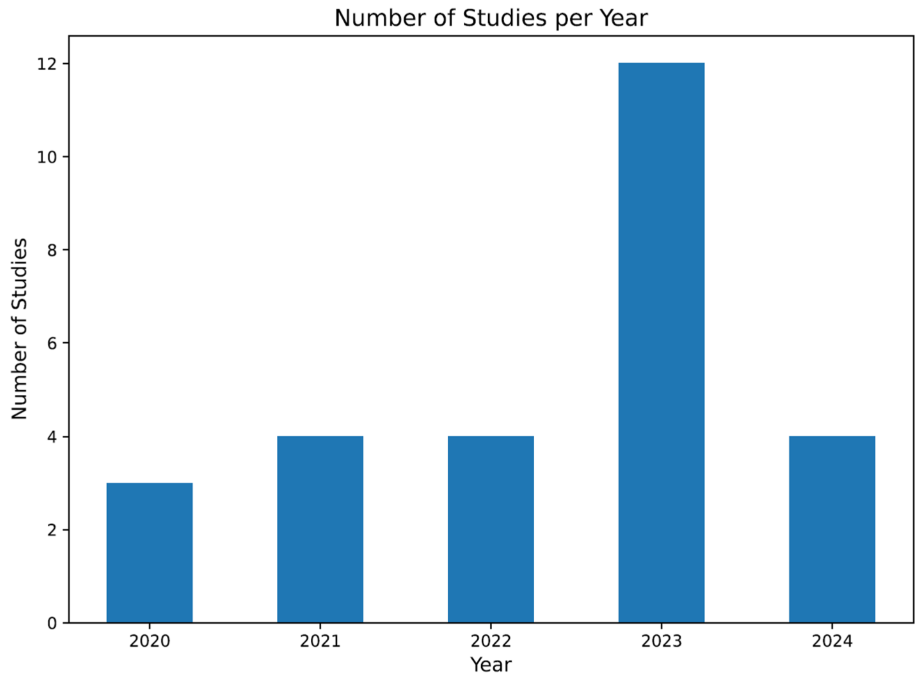

Fig. 2 Distribution of selected studies by publication year

objectives. A standardized data extraction form was utilized to ensure consistency and comprehensiveness.

The following data points were extracted from each study:

- *Techniques and Methods*: Algorithms and approaches used for audio deepfake detection.
- *Features*: Audio and acoustic features leveraged for detection systems.
- *Performance Metrics*: Detection rates, accuracy, and other evaluation parameters.
- *Datasets*: Publicly available or proprietary datasets used for training and testing models in the studies.

2.2.1 Duplicate removal and bias handling

Standardized keyword combinations are used across all databases to avoid search bias. The same inclusion and exclusion criteria are applied across databases to minimize screening bias. The string was only adjusted to support indexing rules. Duplicate entries were removed using a reference manager, and manual verification was also performed to prevent repetition and redundancy. Uniform criteria are applied across all databases for a consistent search. The summary of the PRISMA process followed is presented in the Table 2.

2.3 Thematic categorization of studies

The extracted data were analyzed to identify recurring themes and patterns. Thematic analysis revealed key areas of focus in the reviewed studies. Table 3 shows the themes and its details. These thematic categories have helped synthesize insights and trends across the selected studies. The PRISMA flow diagram in Fig. 3 provides a visual summary of the study selection process. It outlines the identification, screening, eligibility, and inclusion stages, detailing the number of studies at each step.

Table 2 PRISMA study selection summary

PRISMA stage	Actions taken	Records found
Identification	Records from 4 databases: IEEE Xplore, ACM Digital Library, SpringerLink, MDPI, and Elsevier. Publications from January 2020 to December 2024.	180 records retrieved
Screening	Automated and manual duplicate removal. Two screening rounds (abstract and full text) were applied using inclusion and exclusion criteria.	50 duplicates removed, 130 records screened, 9 records not retrievable
Eligibility	Full-text review of remaining studies against predefined criteria.	121 full-text studies assessed. Excluded: 70 irrelevant abstracts/chapters, 10 video/image-only, 14 review papers
Inclusion	Studies meeting all criteria included in the final SLR synthesis.	27 studies included

Table 3 Thematic categorization

Theme	Details
Detection Techniques	Algorithms and methods developed for identifying audio deepfakes include GANs, CNNs, and RNNs.
Acoustic Features	Specific features such as pitch, timbre, and mel-frequency coefficients are used in detection.
Performance Metrics	Metrics used to evaluate detection systems include accuracy, precision, and recall.
Datasets	Benchmark datasets, such as ASVspoof and FakeAVCeleb, are utilized for training and validation.
Challenges, Gaps	and Limitations in existing methods, as well as future research directions.

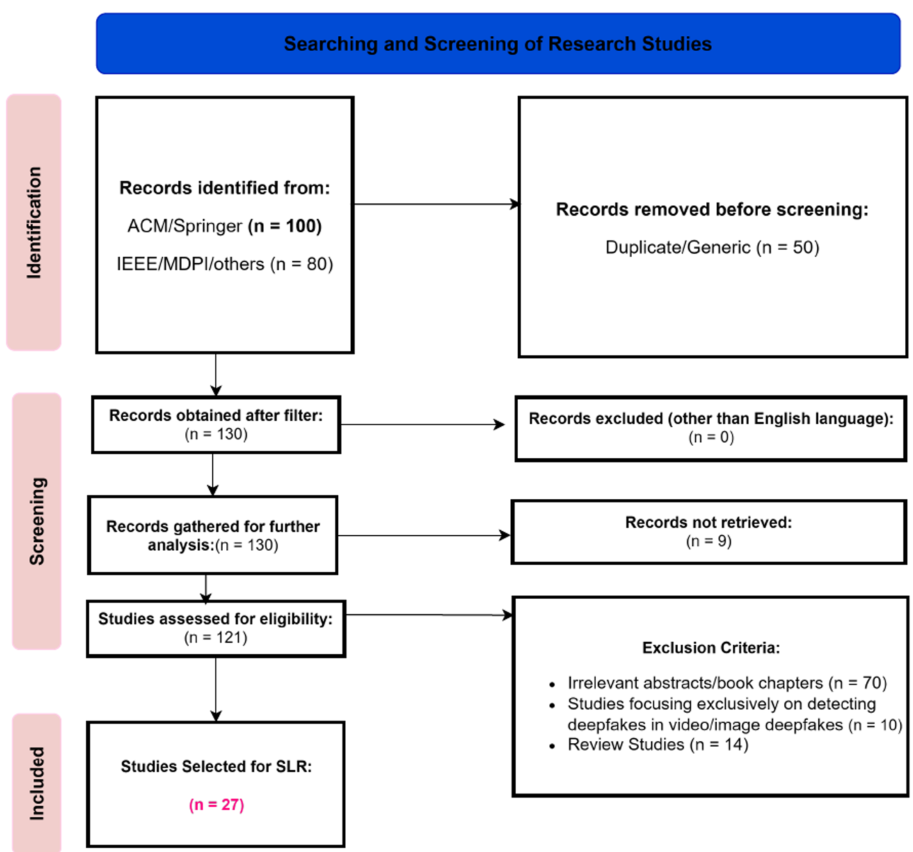

Fig. 3 PRISMA diagram for study selection

3 Analysis of literature

Several authors have proposed novel frameworks to detect audio deepfakes in various scenarios. The study [34] introduces a novel audio-visual deepfake detection method by using affective cues. The authors proposed a Siamese network architecture that uses triplet loss to model the similarity between visual and audio features. They extracted and analyzed the affective cues, such as perceived emotions, to enhance detection accuracy. The features included 13 MFCC for speech through pyAudioAnalysis. The system was tested on the DF-TIMIT and DFDC datasets, which included audio-visual modalities. The method achieved a per-video AUC of 84.4% on DFDC and an AUC of 96.6% on the DF-TIMIT dataset, which was comparable to top benchmarks. Study [35] proposed a novel deep ensemble learning framework called DeepfakeStack. It was designed to enhance the detection of deepfakes by combining multiple DL classifiers. Authors addressed the limitations of individual classifiers, such as overfitting and limited generalizability, and utilized a stacking ensemble approach. The base learners included state-of-the-art CNN architectures, such as XceptionNet, ResNet101, DenseNet121, and MobileNet. Models were trained on the FaceForensics + + dataset that contained 1000 real multimedia content and their manipulated counterparts created with techniques like Deepfake, FaceSwap, and Face2Face. The ensemble model has achieved an accuracy of 99.65% and an AUROC of 1.0 and outperformed all individual models. In the study [36], the authors used a Vision Transformer model to detect deepfakes by integrating it with a distillation methodology. The architecture integrated patch embeddings from Vision Transformers with CNN-derived features. The evaluation metrics, such as Area Under the Curve (AUC) and F1-score, were used, and model achieved an AUC of 0.978 and FI score of 91.9% on the DFDC dataset with over 100,000 multimedia files.

Another study [37] presented an approach to detect deepfakes by jointly analyzing video and audio modalities. The study addressed the synchronization inconsistencies between visemes and phonemes. Study identified the critical challenge of detecting cross-modal manipulations. The authors have proposed a two-plus-one-stream network architecture. This method has introduced a synchronization stream to detect temporal alignment discrepancies between the modalities. They created a custom dataset by combining existing video datasets (FaceForensics + + and DFDC) with vocoder-manipulated audio. The model has achieved up to 97.62% whole-video accuracy on the DFDC dataset. This framework significantly improved generalization to unseen categories, which has made it a valuable advancement in deepfake detection research. The authors in the study [38] have also addressed the limitations of existing deepfake datasets that lacked synchronized and labeled multimodal datasets. To fill this gap, the authors have introduced FakeAVCeleb, a large-scale dataset that contains over 20,000 video-audio samples. The dataset has used state-of-the-art generation techniques for both modalities. FaceSwap and FSGAN for face-swapping, along with Wav2Lip for audio-driven facial reenactment, and SV2TTS for real-time voice cloning. Models such as Xception and MesoNet performed poorly, with average AUCs around 65% and demonstrated the complexity of the dataset. Another study [39] addressed the growing challenge of detecting multi-modal deepfakes. Authors have proposed AVTENet, that was an audio-visual transformer-based ensemble network. AVTENet integrated three sub-networks, such as a video-only network (VN), an audio-only network (AN), and an audio-visual network (AVN). The FakeAVCeleb dataset, generated through manipulation techniques like FaceSwap, FSGAN, Wav2Lip, and Real-Time Voice Cloning (RTVC) was used in evaluation. The model used mel-spectrograms for audio features, ViViT for visual token encoding, and AV-HuBERT for joint embeddings. Performance metrics such as accuracy, precision, recall, and F1-score were used, with AVTENetff, a feature fusion network, achieved 99% accuracy in comparison with existing CNN-based methods.

The study [40] has addressed a major gap in audio deepfakes datasets. The authors have introduced a large-scale dataset called WaveFake that was specifically developed for audio deepfake detection. The dataset included 117,985 generated audio clips that collectively made up approximately 196 h of data. Dataset was developed using six state-of-the-art generative models such as MelGAN, HiFi-GAN, and WaveGlow. The study systematically analyzed the spectral features (Mel-spectrogram, MFCC) of these synthetic audio samples and identified subtle and model-specific differences such as particularly in high-frequency regions. To benchmark detection capabilities, two classifiers were implemented, a Gaussian Mixture Model (GMM) that uses Linear Frequency Cepstral Coefficients (LFCC) and a neural network-based classifier, RawNet2, that was designed for end-to-end detection. Evaluations were done on metrics such as the Equal Error Rate (EER) and GMM. EER values ranged from 0.008 (ideal) to 0.533, which showed poorer performance in unseen data. The dataset is derived primarily from LJSpeech and JSUT corpora to ensure diversity in language and speaker characteristics. The study [2] introduced the ADD 2022 challenge that aimed at advancing audio deepfake detection methods through realistic and challenging scenarios. The challenge focused on Low-Quality Fake Detection (LF), Partially Fake Detection (PF), and the Audio Fake Game (FG). These tracks addressed key issues such as identifying fully synthesized audio with added noise as well as detecting partial manipulations in real speech. The authors have used datasets for ADD 2022 that included samples derived from AISHELL-1, AISHELL-3, and AISHELL-4 corpora and featured diverse Mandarin speech. The dataset is split into training, development, adaptation, and test sets, ensuring no overlap in speakers across these subsets. Evaluation metrics included EER for detection tasks and Deception Success Rate (DSR) for adversarial generation. Results highlighted that the LF track's difficulty, with an average EER of 31.7%, while the best submission achieved 21.7%. For PF detection, the average EER improved to 37.9%, with a best result of 4.8%. Study [41] extended the ADD 2022 challenge with ADD 2023 by introducing additional tasks such as region localization and algorithm recognition. The iteration introduced three subchallenges such as Audio Fake Game (FG), Manipulation Region Location (RL), and Deepfake Algorithm Recognition (AR). These tasks moved beyond binary classification to localizing manipulated segments in partially fake audio and identifying the algorithm responsible for generating fake audio. The datasets for ADD 2023 were also developed using corpora like AISHELL-1, AISHELL-3, and THCHS30, with new categories in AR tracks. Evaluation metrics were selected based on each track such as Deception Success Rate (DSR) for FG, Weighted Equal Error Rate (WEER) for FG-D. Baseline models included GMM, LCNN, and ResNet-18, with RL achieving an average score of 48.82% and AR achieving 89.63%.

Authors in the study [42] have explored the application of unsupervised pretraining models for detecting fake audio in the ADD 2022 challenge. The study focused on low-quality (Track 1) and partially fake audio (Track 2). Study used XLS-R, a Wav2Vec-style model that was pretrained on 436k hours of unlabeled speech. The authors addressed the challenges posed by mismatched and noisy datasets. For Track 1, it was observed that XLS-R, which was fine-tuned with labeled data, achieved an EER of 0.33% on the adaptation dataset. However, model struggled with unseen evaluation data and achieved an EER of 34.02%. This highlights challenges in generalizing to diverse audio environments. For Track 2, the authors simulated partially fake data by inserting synthesized or real clips into genuine notes. They have again used XLS-R with fine-tuning, and the model achieved the best EER of 4.80%. Another study [43] has focused on improving the generalization capability of audio deepfake detection systems. Authors have addressed challenges posed by the progressive spoofing techniques in TTS and VC technologies. The authors have proposed a novel framework that has combined Large Margin Cosine Loss (LMCL) and frequency masking augmentation (FreqAugment) to enhance robustness against unknown spoofing attacks. The evaluation was conducted on the ASVspoof 2019 Logical Access (LA) dataset, and the authors also created augmented datasets with real-world distortions. The baseline ResNet-18 model achieved an EER of 4.04%, and by incorporating LMCL and FreqAugment, it was reduced to 1.26%. Other than EER, the Tandem Detection Cost Function (t-DCF) was also used with a lower t-DCF.

Another study [44] has addressed the challenge of detecting audio deepfakes generated by modern TTS systems through a dual approach using feature-based and image-based classification methods. The study used Fake or Real (FoR) dataset, which included 195,000 audio samples from both real human speech and synthetic speech created by advanced algorithms. The feature-based classification method extracted 37 spectral features, such as MFCCs, spectral centroid, and chroma features, which are processed using ML models like Support Vector Machines (SVM), Random Forest, and XGBoost. All of these models achieved relatively better test accuracies, and SVM showed the best performance at 67%. The image-based approach has used melspectrograms of audio data as input to DL models like Spatial Transformer Network (STN) and Temporal Convolutional Network (TCN). The TCN outperformed all other models by achieving a 92% test accuracy and showed that it's a computationally efficient and accurate framework for detecting audio deepfakes. Another study [45] presented a dataset called LAV-DF for audio-visual forgery detection. The LAV-DF dataset was developed with with strategically modified segments using advanced tools like SV2TTS for audio generation. This enabled detection of subtle manipulations often missed by binary classifiers. The authors proposed the Boundary Aware Temporal Forgery Detection+ (BA-TFD+) that has combined Multi-scale Vision Transformers (MViT) for visual frames and ViT-based encoders for audio spectrograms. It was trained using contrastive and multimodal boundary matching losses. The method was evaluated against benchmarks like DFDC and ForgeryNet, and BA-TFD + has achieved 96.3% AP@0.5 for temporal localization on LAV-DF.

The study in [46] proposed Multimodaltrace for audio-visual deepfake detection. The proposed model uses joint representation learning to address synchronization inconsistencies in manipulated media. Authors argued that previous methods treated audio and video separately and this framework integrates modalities through IntrAmodality Mixer Layers (IAML) and IntErmodality Mixer Layers (IEML). The model has extracted features using ResNet-1D for audio and ResNet-3D for video and captured spectral and spatiotemporal characteristics. The study evaluates Multimodaltrace on the FakeAVCeleb dataset, and the model has shown 92.9%. Cross-dataset evaluations on the World

Leaders Dataset (WLD) and Presidential Deepfake Dataset (PDD) demonstrated generalizability, with accuracies of 91.66% and 70%, respectively. Another study [47] has introduced an Audio-Visual Temporal Synchronization for Deepfake Detection (AVTS-DFD). The approach used self-supervised learning to pretrain temporal feature extractors for audio-visual synchronization to overcome the need for large labeled datasets. The model has incorporated contrastive loss (InfoNCE) during pretraining and learned robust embeddings that captured the temporal coherence between lip movements and speech. The method was evaluated on FakeAVCeleb, FaceForensics++, and DFDC datasets and achieved 94.4% accuracy on FakeAVCeleb and showed better results in cross-dataset generalization. Its focus on temporal synchronization enabled strong cross-manipulation performance with average AUC of 98%. Another study [48] has proposed a unified DL framework for detecting deepfakes across both audio and visual modalities in multimedia content. They proposed AVFakeNet and addressed limitations in existing models by introducing a Dense Swin Transformer Net (DST-Net) architecture. The transformer framework has combined dense layers for detailed feature extraction and a modified Swin Transformer for global and local spatio-temporal analysis. The framework was evaluated on multiple datasets, such as FakeAVCeleb, ASVSpoof-2019, Celeb-DF, and cross-dataset scenarios with the World Leader Dataset and the Presidential Deepfakes Dataset. The model achieved 97.51% accuracy on Mel-Spectrogram-based audio evaluation and 92.59% on multimodal detection on FakeAVCeleb.

Authors in the study [49] have developed CFAD, that was a public Chinese dataset. The dataset was designed to address the lack of diverse, real-world audio deepfake benchmarks. The dataset consisted of 12 fake audio types, like traditional vocoder-based systems such as STRAIGHT, Griffin-Lim, and neural vocoder systems like LPCNet and HifiGAN, along with partially fake audio created via splicing. Real audio samples were sourced from six corpora in order to simulate realistic scenarios. LFCC-GMM, LFCC-LCNN, and RawNet2 were evaluated, in which the LFCC-LCNN model was trained on clean data. The model has achieved an EER of 1.26% on clean and known test data but struggled with unseen noisy and codec conditions. CFAD's structure achieved 97.26% F1-score when tested on known fake types under clean conditions.

Another study [50] has proposed the Multi-Feature Audio Authenticity Network (MFAAN) to address the issues of audio deepfakes. The model used a multi-path architecture processing three audio features, such as MFCCs for timbral texture, LFCCs for linear spectral analysis, and Chroma-STFT for harmonic content through dedicated 1D convolutional layers. Authors have evaluated their model on the In-the-Wild Audio Deepfake Dataset and the Fake or Real Dataset. The model achieved an accuracy of 98.93% and 94.47%, respectively, with 0.04% EER on In-the-Wild data. The study [51] explored the efficacy of different CNN-based architectures for detecting deepfake audio in forensic investigations. A study has evaluated MFCC, Mel-spectrum, Chroma- gram, and Spectrogram features. Study demonstrates that custom CNN model achieved highest performance on Chromagram features with an accuracy of 83.6%, VGG-16 outperformed other models when using MFCC features with an accuracy of 86.9%. Another study [52] has performed an investigation on the comparative capabilities of humans and AI algorithms in detecting audio deepfakes by using a gamified experimental framework. The study involved 472 participants competing against AI to classify audio from the ASVspoof 2019 dataset, which consisted of bonafide and spoofed audio generated using TTS, VC, and waveform synthesis methods. The AI models included RawNet2 and a baseline BiLSTM, which used spectrograms. AI model slightly outperformed humans participants with 95% accuracy versus human performance of 80%. Under realistic conditions, both human and AI struggled with advanced attacks such as Tacotron 2 and Wavenet. They achieved similar accuracies of 50–60. The study [53] proposed SpecRNet, which was a spectrogram-based NN designed for audio deepfake detection. The study focused on reducing model complexity compared to RawNet2 and LCNN. The model processed LFCCs as input and used a lightweight architecture with residual blocks and bidirectional GRU layers for temporal analysis. Model was evaluated on on the WaveFake dataset that has 104,000 fake samples generated using eight advanced TTS and VC methods The model achieved an EER of 0.1549 and an AUC of 99.99 while requiring 40% less inference time than LCNN. Authors in the study [54] have proposed a CNN-based framework for detecting deepfake audio and used MFCC for feature extraction. Their approach used MFCC's to capture timbral and tonal characteristics indicative of synthetic speech. The model uses 1D CNN layers to extract temporal patterns followed by ReLU activations max-pooling, and fully connected layers for classification. The dataset consisted of real and fake audio samples collected from diverse sources, such as public speeches and audiobooks, and had variations in gender, accents, and background noise. The model achieved an accuracy of 84% and an F1-score of 0.998 when evaluated on metrics against various manipulations. Authors in [55] have proposed a multimodal deepfake detection method through extensive classification and binary detection. The model identified inconsistencies in four distinct types of media including real video-real audio, real video-fake audio, fake video-real audio, and fake video-fake audio. Feature extraction has used Capsule Networks and Swin Transformers for visual and audio modalities and used face detection and mel-spectrogram processing for input preparation. The proposed system was evaluated on FakeAVCeleb and TMC datasets, and under intra-domain testing, the model has achieved an AUC of 99.30% and accuracy of 99.20% on FakeAVCeleb. Cross-domain performance also demonstrated strong generalizability, with the Swin Transformer variant achieving an AUC of 82.17% when trained on TMC and tested on FakeAVCeleb. Another study [56] has proposed a framework called AVA-CL to detect audio deepfakes. The model used audio-visual attention and contrastive learning to enhance detection accuracy. The model processed video frames and audio streams using MARLIN for facial feature extraction and MFCC for audio, followed by temporal alignment through 1D convolutional layers. A fusion mechanism integrated audio-visual features, while InfoNCE loss narrowed the semantic gap by pulling positive pairs closer and pushing negative ones further away. The model has achieved an AUC value of 88.64% on DFDC, 89.47% on FakeAVCeleb, and 99.99% on DeepfakeTIMIT. A study [26] developed Arabic-AD that was a self-supervised DL framework designed to detect both imitation-based and synthetic Arabic deepfake audio. The authors addressed the gap in Arabic speech deepfake detection. They have developed three specialized models by utilizing datasets of Modern Standard Arabic (MSA) and accented Arabic speech from both native and non-native speakers. Key datasets included Ar-DAD for imitation fakes and newly generated synthetic datasets using FastSpeech2. The Arabic-AD framework has integrated HuBERT (Hidden-Unit Bidirectional Encoder Representations from Transformers) for feature extraction and classification. Models achieved exceptional performance with 97% accuracy and an EER of 0.027% for imitation fakes.

Authors in ref. [57] have introduced AVForensics, which was a self-supervised, audio-driven, multimodal framework. The framework exploited audio-visual correlations to overcome the limitations of mouth-region-only detectors. AVForensics used spatio-temporal transformers (STTr) for video encoding, complemented by frequency-aware features to enhance resilience against compression. The model was pretrained on the VoxCeleb2 dataset and achieved significant generalization on deepfake datasets such as FaceForensics++, CelebDFv2, and DFDC, with AUC scores of 99.9%. This study [58] has investigated the application of MFCC combined with ML models to detect audio deepfakes. The study emphasized that audio deepfakes in digital forensics have shown the potential for misuse in identity theft, scams, and cybercrime. Authors have used the Fake-or-Real dataset and structured it to include sub-datasets such as for-original and for-rerec, that has captured both original and re-recorded audio with variations in quality and length. The study applied SVM, Random Forest (RF), and VGG-16 to classify audio based on MFCC features. Evaluations have been performed using metrics like accuracy, F1-score, recall, etc. SVM has outperformed other models by achieving a detection accuracy of 98.83% on the for-research sub-dataset, while the VGG-16 model demonstrated better results with an accuracy of 93% on the for-original sub-dataset. The study focused on MFCC-based feature extraction and preprocessing to capture spectral and timbral cues. Noise-handling mechanisms are also implemented to improve the robustness of the detection system in real-world forensic scenarios.

Table 4 presents the summary of studies discussed in the literature analysis.

4 Results and analysis

Building on the diverse methodological contributions outlined in the literature, this section presents the findings and further provides answers to the devised RQ, stated in Sect. 2.1.

4.1 RQ1: what are the existing techniques and algorithms discussed on audio deepfake detection?

The field of audio deepfake detection has advanced significantly, and literature has covered a range of techniques from traditional feature-based approaches to novel deep learning frameworks. One of the widely adopted techniques is the Feature-Based Technique. It is a foundational approach that involved extracting features from audio signals for classification. Several studies [44, 49] have utilized MFCC and LFCC as key spectral features. These features have captured the tonal and timbral aspects of audio, which are critical for distinguishing between real and synthetic speech. Algorithms like GMM and traditional SVM operated on these features to identify patterns indicative of manipulation. For example, studies [34, 35, 40] use GMM and LFCC, while study [58] relied on MFCC and SVM for feature-based classification. These techniques focused on analyzing specific artifacts introduced during audio generation, often targeting high-frequency regions or subtle inconsistencies.

Several studies [43, 51, 53] have extended feature-based approaches by utilizing DL and CNNs to automatically learn features from spectrograms, a visual representation of audio signals. End-to-end frameworks directly process raw audio signals without relying on manual feature extraction. Techniques like RawNet2 and SpecRNet used deep neural networks with residual blocks and GRU layers to model spectral and temporal

Table 4 Summary of studies discussed in the literature analysis

Studies	Year	Issue found	Technique used	Algorithm
[48]	2023	Unified audio-visual deepfake detection, addressing manipulations in audio streams.	Dense Swin Transformer Net (DST-Net)	DST-Net
[51]	2023	Detection of deepfake audio.	Use of CNN architectures (Chromagram, Mel-Spectrum, MFCC, Spectrogram) for feature extraction.	VGG-16, FG-LCNN, ResNet, Custom CNN architecture
[35]	2020	Detection of manipulated deepfakes	Deep Ensemble Learning (DeepfakeStack) multiple DL models	XceptionNet, ResNet101, InceptionV3, DenseNet121, 169, MobileNet, InceptionResNetV2, (DFC)
[34]	2020	Audio-visual deepfake detection using perceived emotions and modality correlations	Multimodal learning combining facial and speech features; Siamese network	OpenFace for face features, pyAudioAnalysis for speech, MFN, Siamese
[36]	2021	Detection of deepfake videos by improving generalization and reducing overfitting	Vision Transformer with distillation methodology, combining CNN features	Vision Transformer, EfficientNet
[52]	2022	Comparison of human and AI detection capabilities for audio deepfakes	Gamified framework for human vs. AI comparison	BiLSTM, RawNet2 (AI models)
[38]	2021	Development of a novel multimodal deepfake dataset	Multimodal dataset (FakeAVCeleb)	Autoencoders, Variational Autoencoders, GANs
[40]	2021	Creation of an audio deepfake detection dataset and evaluation of detection methods	Frequency analysis and prosody analysis of deepfake audio	Gaussian Mixture Model RawNet2, MFCC, LFCC
[41]	2023	Detection of audio deepfakes and manipulation region localization	GMM, LFCC, LCNN	GMM, LCNN, ResNet
[2]	2024	Detection of low-quality and partially fake audio in real-world scenarios	GMM, LCNN, RawNet2 applied to low-quality, partially fake, and audio game scenario	GMM, (LCNN), RawNet2
[53]	2022	Developing a faster, accessible, and efficient audio deepfake detection model	Spectrogram-based processing using a lightweight architecture	SpecRNet, LCNN, RawNet2 focusing on reduced computational complexity
[49]	2024	Creation of the CFAD dataset for fake audio detection under noisy and codec conditions	LFCC features with GMM and LCNN, RawNet2	LFCC-GMM, LCNN, RawNet2
[45]	2023	Audio-visual forgery detection and temporal localization	Boundary Aware Temporal Forgery Detection	Multiscale Vision Transformer, 3D CNN
[55]	2023	Improving detection of multimodal (audio-visual) deepfakes	Fine-grained deepfake identification, integrating audio-visual data	Capsule Network, Swin Transformer
[39]	2023	Multi-modal deepfake detection leveraging audio and visual cues	AVTENet	Audio-Visual Transformer, AST, ViViT, AV-HuBERT
[50]	2023	Audio deepfake detection using multiple feature extraction techniques	MFAAN combining MFCC, LFCC, Chroma-STFT	CNN, MFCC, LFCC, Chroma-STFT
[57]	2023	Detection of deepfake videos using audio-visual matching with full face features	AVForensics	Memory-efficient Transformer, Masking Strategy
[56]	2024	Improving robustness and generalization in audio-visual deepfake detection	AVA-CL model	Memory-efficient Transformer, InfoNCE loss
[43]	2020	Generalization in audio deepfake detection for logical access attacks	LMCL, frequency masking augmentation	ResNet18, ResNet18-L, ResNet18-L-FM

Table 4 (continued)

Studies	Year	Issue found	Technique used	Algorithm
[54]	2024	Detection of deepfake audio using MFCC feature extraction techniques	MFCC for feature extraction combined with CNN for audio classification	CNN, Fully Connected Layers
[59]	2023	Detecting fake audio for Arabic speakers using self-supervised deep learning	Arabic-AD method using SSL with MSA dataset	HuBERT, GMM, CNN, LSTM
[42]	2022	Detection of low-quality and partially fake audio using unsupervised pre-training models	XLS-R model for feature extraction	XLS-R, ECAPA-TDNN
[47]	2023	Audio-visual deepfake detection using temporal synchronization	Self-supervised learning for audio-visual temporal synchronization	AVTS-DFD, ResNet-18, VGG-16
[46]	2023	Detection of deepfakes using joint audiovisual representation learning	Multimodaltrace framework using IntrAmodality and IntErModality mixing layers	ResNet-3D (visual), ResNet-1D (audio)
[44]	2022	Detection of audio deepfakes using feature-based and image-based approaches	MFCC Image-based classification	SVM, TCN, STN, CNN, Random Forest, XGBoost
[37]	2021	Joint detection of audio-visual deepfakes using synchronization	Two-plus-one stream model with synchronization-based learning	Late-fused joint prediction, Inter-attention mechanism

patterns in audio. These models eliminated the preprocessing overhead while maintaining the ability to capture subtle manipulations in raw waveforms. Techniques like Custom CNNs, ResNet, VGG-16 were used in studies [35, 38, 54]. These networks were designed to process spectrogram inputs, enabling the detection of subtle patterns and inconsistencies that are not easily captured by handcrafted features. Ensemble approaches, such as those combining multiple CNN architectures, further enhanced detection by aggregating diverse model perspectives.

Another notable advancement has been observed towards self-supervised learning in audio deepfake detection. Techniques like HuBERT and XLS-R allowed for pretraining on large-scale unlabeled datasets. Studies [26, 41, 42, 47, 57] have utilized these frameworks to generate robust embeddings that encapsulate both semantic and acoustic properties of audio. These embeddings were then fine-tuned for downstream classification tasks that have provided a generalizable and noise-resilient approach to deepfake detection. Along with that, Transformer-based architectures have gained prominence for their ability to model long-term dependencies in data. Studies [36, 39, 41, 48, 57] proposed architectures such as ViT and DST-Net for processing audio signals. These models often integrated transformers with CNNs to combine local feature extraction with global contextual understanding. Techniques like patch embeddings and hierarchical token processing enhanced the detection of subtle artifacts in synthetic audio.

Several studies have addressed the limitations of single-modality models by introducing multimodal frameworks that analyzed audio-visual data. Techniques such as synchronization stream networks [37] and audio-visual transformers [39, 45] were used to detect cross-modal inconsistencies. These methods integrated audio features like MFCCs and mel-spectrograms with video features such as lip movements, enabling the detection of manipulation in both modalities. By exploiting interdependencies, these approaches identified synchronization anomalies and other signs of tampering.

To address environmental challenges like noise and codec distortions, studies [43, 49] incorporated augmentation techniques such as frequency masking (FreqAug) and SpecAug. These methods introduced variability in the training data by improving the robustness of models against real-world audio conditions. Models like ResNet were optimized with large-margin cosine loss to enhance resilience against unknown spoofing attacks. In addition to that, ensemble methods combine multiple base learners to improve detection performance. DeepfakeStack [35] used a stacking ensemble of CNNs like XceptionNet and ResNet101, while AVTENet [39] incorporated transformer-based sub-networks for audio and video processing. These ensemble approaches aimed to reduce overfitting and increase generalizability by aggregating the strengths of individual models. This has given rise to the hybrid feature processing techniques.

These techniques have provided a balanced approach by combining traditional features and learned representations. For instance, the Vision Transformer model [36] integrated patch embeddings from transformers with CNN-derived features and ensured a robust analysis of both local and global patterns in audio. Contrastive learning techniques, as discussed in ref. [56] used InfoNCE loss to narrow the semantic gap between real and fake audio by maximizing agreement between positive pairs and minimizing agreement between negative pairs. This approach enhanced the robustness of multimodal models by aligning audio and video features in a shared latent space and effectively addressed synchronization anomalies.

Figure 4 shows the frequency of techniques discussed in the literature.

4.2 RQ2: what are the current features used in audio deepfake studies?

The current features used in audio deepfake detection include a range of spectral, temporal, and learned representations that address the complexity of synthetic audio manipulation. The literature has covered these features as presented in Table 5.

As seen in Table 5, MFCC has been known for its robustness in modeling the spectral envelope of speech, aligning closely with human auditory perception. This has made it a valuable indicator for detecting tonal inconsistencies introduced by generative models. Studies [34, 35, 38, 50, 58] have utilized MFCC to good effect in identifying subtle

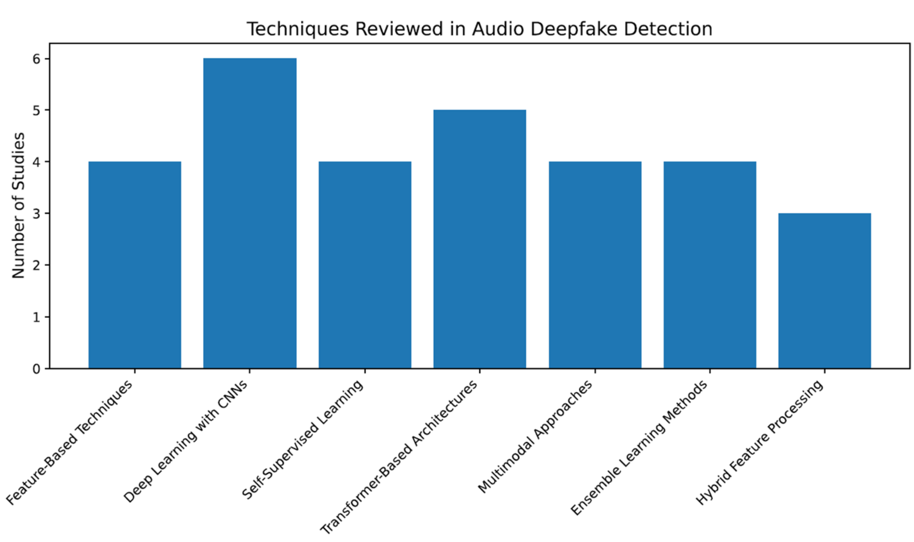

Fig. 4 Techniques reviewed in audio deepfake detection

Table 5 Features used in audio deepfake detection

Feature type	Description	Studies
MFCC (Mel-Frequency Cepstral Coefficients)	Represents short-term power spectrum of audio, widely used for its ability to capture tonal characteristics and mimic human auditory perception.	[34, 35, 38, 50, 58]
LFCC (Linear Frequency Cepstral Coefficients)	Extracts linear spectral information, particularly effective for high-frequency regions to capture artefacts of synthetic audio generation.	[40, 42, 49]
Mel-Spectrogram	Converts audio signals into a visual time-frequency representation, suitable for deep learning models such as CNNs and transformers.	[38–40, 45]
Chromagram	Represents harmonic and pitch-based characteristics, highlighting tonal content in the audio.	[51, 54]
Delta and Delta-Delta Coefficients	Captures the temporal evolution of MFCC features, adding information about changes in frequency content over time.	[35, 50]
Raw Audio Features	Directly processes raw audio waveforms without manual feature extraction, allowing models to learn patterns directly from the signal.	[40, 53]
Spectral Centroid	Measures the "center of mass" of the spectrum, indicating where most of the energy in the signal is concentrated.	[44]
Temporal Features	Analyzes synchronization between audio and visual modalities, focusing on lip-speech alignment and temporal inconsistencies.	[37, 39, 45]
Filter Bank Features	Captures band-limited frequency responses, providing a compact representation for neural network input.	[42, 43, 50]
Log Filter Bank Energy (LFBE)	A logarithmic transformation of filter bank energy, often used in conjunction with spectrograms for better model accuracy.	[40, 49]
Chroma-STFT (Short-Time Fourier Transform)	Combines harmonic structure with temporal features for deep learning frameworks to detect forged audio patterns.	[38, 50]
Wavelet Transform Features	Captures frequency and time domain information simultaneously, suitable for analyzing transient and steady-state components of audio.	[35, 50]
Learned Embeddings (e.g., XLS-R, HuBERT)	High-dimensional representations generated by self-supervised learning models, encapsulating both acoustic and semantic features.	[26, 42, 57]
Synchronization Features	Detects alignment anomalies between lip movements and speech sounds in multimodal settings, indicating cross-modal tampering.	[37, 39, 45]

timbral deviations. Delta and delta-delta coefficients extended MFCC by incorporating temporal derivatives, and these were particularly useful for detecting abrupt transitions or temporal anomalies, and have been used in refs. [35, 50]. LFCC as discussed in refs. [40, 42, 49] has also been significant in capturing linear spectral details that were particularly useful in exposing high-frequency noise patterns that are often artifacts of synthesis.

As deep learning frameworks have been adopted widely, more advanced time-frequency representations such as mel-spectrograms and chromagrams were used. Mel-spectrograms' 2D image-like format has enabled CNNs and transformers to perform visual analysis on audio content. This representation has become fundamental to deep learning models such as SpecRNet and AVTENet, as shown in studies [38–40, 45]. Chromagrams provided a quality for detecting pitch artifacts in tampered audio [51, 54].

Features such as spectral centroid [33] and multimodal frameworks were also used to detect misalignments in lip movements and speech timing in studies [37, 39, 45]. The application LFBE has further allowed for more compact and noise-resilient audio representations. These have been implemented in neural network-based models for improved detection performance, as seen in refs. [40, 42, 43, 49, 50]. Use of learned embeddings from self-supervised models like XLS-R and HuBERT has also been observed. These high-dimensional representations encode both semantic and acoustic information. Their application in audio deepfake detection is rising due to their strong generalization across datasets, as demonstrated in refs. [26, 42, 59].

4.3 RQ3: what are the audio deepfake indicators discussed in existing studies?

Literature has discussed several audio deepfake indicators. These are specific artifacts or inconsistencies that reveal the presence of synthetic or manipulated audio. These include spectral artifacts like high-frequency noise and spectral discontinuities, temporal anomalies such as inconsistent dynamics and synchronization gaps, and phonetic irregularities like unnatural transitions. Analysis demonstrated that, one of the most prominent cues was high-frequency noise, which has often been introduced by synthesis processes that failed to replicate the subtle tonal range of human speech. Studies [35, 40, 49] have used LFCC features to expose these anomalies. Along with that, abrupt, unnatural shifts in the frequency domain have been detected in mel-spectrograms and MFCCs [34, 38, 58]. Irregular transitions in energy across frames have been found to be another common indicator. These were typically identified through delta and delta-delta coefficients, as mentioned in refs. [35, 50, 54]. Several other indicators included monotonic flow as in [26, 42, 57], and the absence of expressive shifts has also been highlighted in refs. [34, 35]. On a phonetic level, unnatural phoneme transitions and harmonic imbalance point toward mismatched synthesis of speech units [38, 44, 50, 58]. Studies also discussed model-specific indicators that were unique to generative architectures such as Wav2Lip and MelGAN [39, 40, 57]. Indicators such as compression issues, poor rendering, or codec mismatches were also discussed in refs. [35, 43, 45, 49, 50]. Table 7 mapped the indicators with performance trends.

4.4 RQ4: what are the current datasets and detection rate in existing studies?

Several datasets have been used by researchers in the literature to achieve diverse accuracies. Table 6 provides the details of the datasets and accuracies achieved by studies in the literature.

The datasets used in audio deepfake detection studies, such as FakeAVCeleb, DFDC, WaveFake, and ASVspoof 2019, have demonstrated diverse accuracies based on the techniques and algorithms applied. Studies achieving over 90% accuracy have predominantly used advanced methods such as transformer-based architectures (e.g., Dense Swin Transformer in ref. [48], ViT in ref. [36] and multimodal approaches combining audio and video modalities [39, 46]. For example, FakeAVCeleb datasets paired with transformer models like AVTENet achieved 99% accuracy, whereas traditional CNN-based approaches such as ResNet showed comparatively lower performance due to their limited ability to capture cross-modal inconsistencies.

However, the accuracy differs significantly across datasets, even with the same techniques. For instance, AVFakeNet achieved 97.51% on FakeAVCeleb but struggled with lower accuracy on cross-dataset scenarios such as ASVspoof 2019, highlighting the importance of dataset characteristics like noise, codecs, and language diversity [48]. Models trained on highly curated datasets like CFAD achieved excellent results under clean conditions but faced challenges in noisy or unseen environments, demonstrating the need for robustness in real-world scenarios [49].

This inconsistency provides empirical evidence to highlight the necessity of designing a technique capable of generalizing across datasets. It can use hybrid models that combine features such as MFCC, LFCC with DL frameworks to enhance robustness and maintain high accuracy rates across diverse conditions. A multimodal and ensemble learning approach could also potentially bridge these gaps by synthesizing the strengths

Table 6 Datasets and accuracies in audio deepfake detection

Dataset	Studies	Technique	Algorithm	Accuracy
FakeAVCeleb (20,000 + samples)	[38]	Multimodal Feature Extraction	MesoNet, XceptionNet	AUC 65%
	[39]	AVTENet	AV-HuBERT, ViViT	99%
	[46]	MultimodalTrace	ResNet-1D, ResNet-3D	92.9%
	[48]	AVFakeNet	Dense Swin Transformer	97.51%, 92.59%
	[47]	AVTS-DFD for Temporal Synchronization	ResNet-18, VGG-16	94.4%
DFDC (100,000 + samples)	[37]	Two-plus-One Synchronization	Convolutional + Sync-Stream Mechanism	97.62%
	[36]	Vision Transformer Hybrid	VT, EfficientNet	97.8%, 91.9%
	[56]	AVA-CL	Memory-Efficient Transformer, InfoNCE Loss	88.64%
WaveFake (117,985 clips)	[40]	Spectral Analysis (LFCC/MFCC)	GMM, RawNet2	EER 0.04
	[53]	Spectrogram-Based Lightweight Model	SpecRNet, LCNN	AUC 99.99%
ASVspoof 2019	[52]	Gamified Human vs. AI Detection	BiLSTM, RawNet2	AI: 95%, Human: 80%
	[43]	FreqAug + LMCL	ResNet18	EER 1.26%
AISHELL (1, 3, 4)	[2]	Low-Quality/Partially Fake Detection	LCNN, GMM	EER: 21.7% (Track 1), 4.8% (Track 2)
	[41]	Region Localization + Fake Game Detection	GMM, LCNN, ResNet	WEER: 12.45%, DSR: 93.8% (Track 3)
Fake-or-Real	[44]	Feature/Image-Based Classification	SVM, TCN, Random Forest, CNN	TCN: 92%
	[50]	Multi-Feature Authenticity Network	CNN + MFCC, LFCC, Chroma-STFT	98.93%
CFAD	[49]	LFCC Feature Analysis	LFCC-GMM, LCNN, RawNet2	F1: 97.26% (Clean), Noisy Challenges
LAV-DF	[45]	Boundary Aware Temporal Detection	Multiscale Vision Transformer	96.3%
VoxCeleb2	[57]	AVForensics Framework	Memory Transformer	98.7%
Public Speeches, Audiobooks	[57]	MFCC Feature Extraction	CNN + Fully Connected Layers	84%
Modern Standard Arabic	[26]	Self-Supervised Arabic Deepfake Detection	HuBERT	97%
Custom Datasets	[35]	DeepfakeStack	XceptionNet, ResNet101	99.65%
	[34]	Affective Cues in Multimodal Learning	MFN, Siamese Network	DFDC AUC 84.4%, DF-TIMIT AUC 96.6%

of individual methodologies and addressing dataset-specific limitations. Figure 5 shows the datasets and the archived accuracy.

4.5 RQ5: what metrics are used to evaluate audio deepfake detection?

Audio deepfake detection studies have utilized a range of metrics to evaluate the methodologies. Accuracy has been widely adopted to measure correct classifications and was used in studies [35, 38, 54, 60]. AUC evaluated the ability to differentiate real and fake audio and was applied in refs. [36, 39]. EER, which balanced false positives and

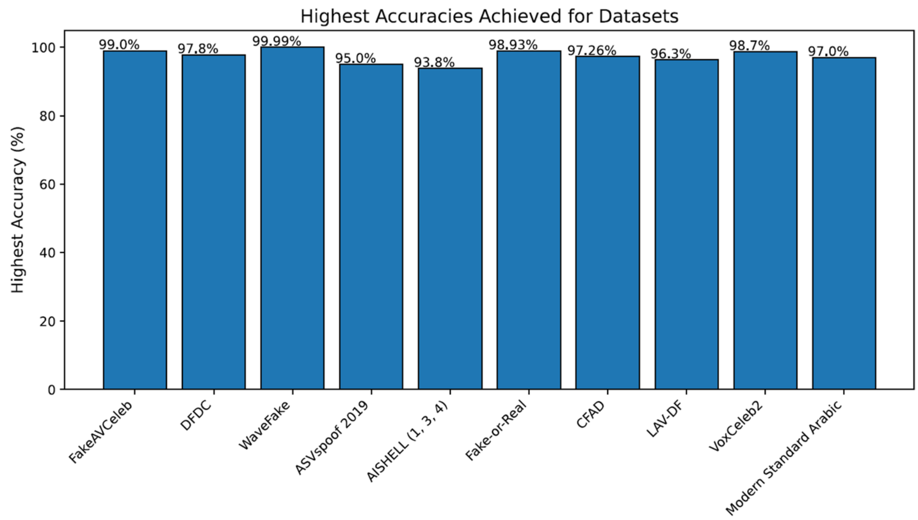

Fig. 5 Highest detection accuracies for major audio and audio-visual deepfake datasets

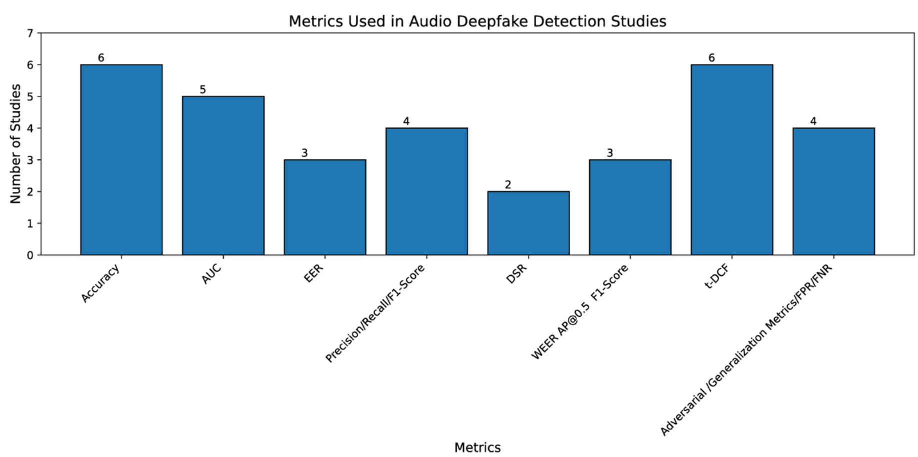

Fig. 6 Metrics reported across audio and audio-visual deepfake detection studies

negatives, was used in refs. [40, 42, 49]. Studies [35, 39] used Precision, Recall, and F1-score to analyze performance. The ADD challenges introduced advanced metrics like DSR and Weighted EER [41]. In addition to that, t-DCF was utilized in refs. [40, 43] for cost-sensitive environments, while cross-dataset generalization metrics evaluated model adaptability [46, 48].

Figure 6 shows various evaluation metrics used by studies in the literature.

As in Fig. 6, it can be seen that accuracy has been the most widely adopted, particularly in controlled settings with balanced datasets. It has been used to demonstrate the effectiveness of CNN-based and ensemble frameworks in correctly distinguishing fake audio from real audio. While high accuracy suggests reliable performance, it may not fully capture model sensitivity in imbalanced or real-world conditions where false positives and negatives carry different weights. To address this limitation, AUC has been used. It is especially useful when the dataset is imbalanced or when different operating points need to be assessed. AUC was used to highlight how well transformer-based models like ViViT or audio-visual fusion networks could separate real and manipulated audio across a wide range of conditions. A high AUC score of 0.978 in ref. [36] indicated strong discriminative power even when raw accuracy may be comparable to other models. EER, as can be seen, has also been used to define and is ideal for evaluating systems where both false positives and false negatives are equally important. Studies [40, 42, 49] utilized EER to benchmark audio deepfake detection on datasets such as WaveFake and CFAD. Lower EERs, such as 0.008, reflected high reliability in detecting synthetic speech, particularly in clean or limited datasets. Several other metrics, such as t-DCF, F1 score, and recall, have been commonly used, but beyond these conventional metrics, DSR and WEER have been used to assess models' resilience to partial manipulations and low-quality spoofing.

4.6 RQ6: what are the challenges and limitations discussed in studies on audio deepfake detection?

Literature has presented several challenges and limitations as discussed in the section.

One significant challenge that was discussed across studies is the generalization capability of detection models when exposed to unseen audio manipulations. Studies [39, 40, 49] emphasized that while detection models performed well on known data, their efficacy diminished with novel synthetic techniques, particularly advanced text-to-speech and voice conversion methods. Another study [52] also stated that detection models often struggled with these sophisticated manipulations. Study [40] noted higher error rates in unseen scenarios, despite achieving low EER in known datasets. Similarly, the author in [49] highlighted that codec distortion and low SNR significantly impacted detection robustness. These findings highlighted the need for models that can reliably adapt to diverse and evolving audio manipulation techniques. As well as achieve better generalization across diverse datasets and languages [54].

Literature has also observed dataset-related issues, including biases, insufficient diversity, and a lack of synchronized multimodal data. Studies [38, 41, 42] pointed out that many existing datasets failed to represent diverse real-world conditions. The study [38] addressed biases by developing the FakeAVCeleb dataset to ensure balanced representations but acknowledged its complexity. The author in ref. [52] has emphasized the scarcity of large-scale datasets for partially fake audio. Without comprehensive datasets including a variety of manipulations and conditions, model training and evaluation remain restricted. Authors in ref. [52] stated that dataset biases are a persistent issue, as models frequently overfit to specific artifacts within curated datasets that limit their ability to perform effectively in diverse, real-world scenarios.

This has resulted in inconsistencies in accuracy across different datasets. Studies [40, 49] highlighted that models often achieved high performance on curated datasets like WaveFake or CFAD under clean conditions but struggled significantly in noisy or unseen environments. For instance, the author in ref. [49] reported an F1-score of 97.26% on clean conditions but noted substantial degradation when tested with codec distortions or low SNR scenarios. Similarly [39], achieved 99% accuracy on the FakeAVCeleb dataset using AVTENet but observed lower performance when tested on cross-domain datasets like ASVspoof 2019. The study [54] also stated that the lack of large-scale, multimodal datasets and standardized evaluation benchmarks is a significant issue.

Another limitation that has been observed was adversarial attacks in audio deepfake detection. Authors in refs. [26, 41, 57] highlighted the vulnerability of detection systems to adversarial noise and attack methods. Study [57] observed that self-supervised frameworks often struggled to maintain performance under adversarial conditions. Another challenge is reliance on features such as MFCCs that limit model adaptability. Studies [35, 50, 54] discussed the constraints of manual feature extraction methods. Study [35] highlighted that while MFCCs capture important tonal characteristics, they lacked robustness when applied to novel scenarios. This dependency on manual features highlights the need for a shift toward end-to-end learning approaches that can automatically learn and adapt to new data.

Another complex challenge that has been observed was the detection of temporal misalignments and unnatural phonetic transitions in synthetic speech. Authors in refs. [34, 37, 39] discussed the difficulty of identifying temporal anomalies between phonemes and visemes in audio-visual deepfakes. Study [34] emphasized the challenge of modeling temporal dependencies in synthetic speech. Several models, when tested in the real world, raised the issue of resource and computational demands. These were critical barriers to the real-world deployment of detection systems. Studies [51, 53, 54] highlighted the trade-offs between accuracy and computational efficiency. Study [53] addressed the issue of computationally intensive models like RawNet2. However, it was observed that the reduced complexity came at the cost of lower performance on complex datasets. Study [54] further emphasized that the high computational requirements of scaling detection models make them impractical for large-scale or real-time applications. In audio deepfake detection, handling noise and real-world diversity has remained a persistent issue. Several studies [42, 43, 49] reported significant performance degradation in noisy or low-SNR environments. Study [43] noted that codec distortions and environmental noise have enhanced the detection challenges, while study [42] observed similar limitations in models trained on clean data. To address these challenges, authors have suggested incorporating noise augmentation techniques during training, but these efforts are still evolving and need more research.

Figure 7 shows the frequency of major challenges identified across reviewed studies, including generalization gaps, dataset constraints and robustness issues.

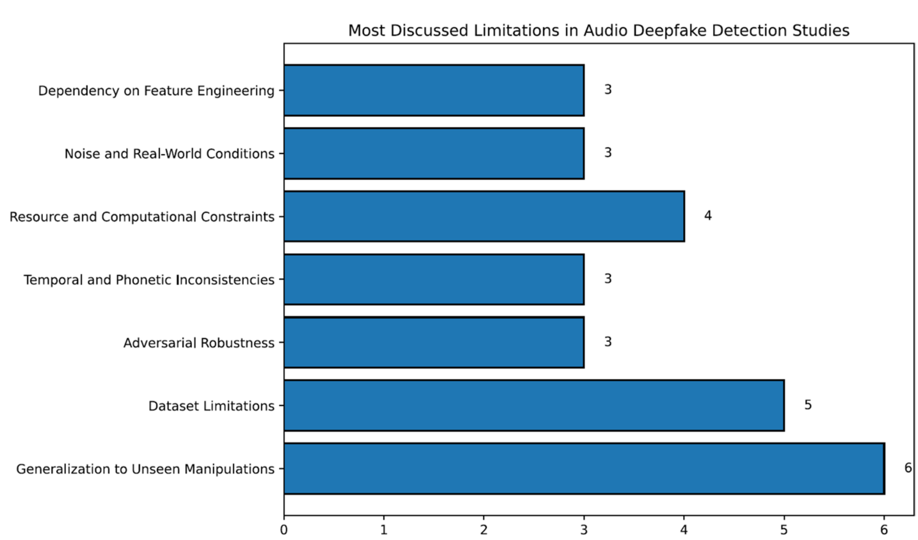

Fig. 7 Key limitations reported in audio deepfake detection literature

4.7 Comparative analysis

Result analysis highlights several clear performance patterns across algorithms and datasets. The studies [40, 43, 50, 51, 53, 54, 58] demonstrate that CNN-based models remain reliable for controlled audio environments and achieve strong results on curated datasets such as WaveFake and CFAD. They also demonstrate that their performance declines when exposed to codec distortions, low-SNR speech, or unfamiliar spoofing techniques. Transformer-based frameworks [36, 45, 48, 57] demonstrate consistently higher accuracy, particularly on large and multimodal datasets such as DFDC and FakeAVCeleb, but, these models introduce computational overhead and show sensitivity to video compression.

Hybrid architectures [36, 37, 46] that combine CNN and Transformer components offer a balanced performance by integrating local spectral cues with long-range contextual patterns. Self-supervised learning approaches [26, 42, 57] show strong generalization across languages and datasets, yet their performance becomes unstable in noisy or mismatched acoustic conditions. The literature highlights that multimodal models outperform conventional models by using cross-modal correlations, and achieve accuracy close to 99%. This advantage comes with increased reliance on high-quality visual input. This highlights a trade-off between accuracy, generalization, and computational practicality. Table 7 presents the comparative analysis of algorithms techniques and datasets mapped according to performance trends.

5 Discussion

Audio deepfakes have already demonstrated their potential for real-world cybercrimes, especially in social engineering and financial fraud. It has been observed how cybercriminals used deepfake audio to impersonate the CEO of a German energy company's parent firm. They mimicked the UK-based CEO's voice so accurately, including accent, tone, and even speech, that the executive believed the request was authentic. This incident exposed a critical vulnerability that even trained professionals can be deceived by well-crafted synthetic audio, especially when delivered in high-pressure or urgent contexts. Traditional voice authentication systems failed in such cases due to their reliance on superficial acoustic features like pitch or cadence, which can now be convincingly replicated using advanced TTS and VC models.

The SLR has comprehensively answered six key research questions that were devised to clarify critical aspects of audio deepfake detection. RQ1 explored existing detection techniques and highlighted the evolution from traditional feature-based methods to advanced DL, multimodal, and transformer-based frameworks. RQ2 analyzed the acoustic and learned features used, such as MFCCs, LFCCs, spectrograms, and self-supervised embeddings like HuBERT and XLS-R. RQ3 identified audio deepfake indicators as well as spectral anomalies, temporal inconsistencies, and model-specific artifacts that were used to detect manipulations. RQ4 synthesized datasets and their corresponding detection rates, offering a unique comparative analysis that revealed how the same technique yielded different accuracies across datasets like FakeAVCeleb, DFDC, and WaveFake. RQ5 evaluated the diversity of performance metrics used, including AUC, EER, F1-score, and DSR, and highlighted limitations in cross-dataset generalization. RQ6 investigated current challenges, such as adversarial vulnerability, noise sensitivity, dataset bias, and computational inefficiency.

The comparative analysis highlights significant advancements and challenges in audio deepfake detection. Some detection frameworks that relied on MFCC or GMM techniques struggled under noisy or real-world conditions, as these features were easily replicated by modern neural vocoders. The consideration of Power Normalized Cepstral Coefficient (PNCC), Constant Q Cepstral Coefficient (CQCC), and Spectral Flux (SF) over MFCC has been mentioned in refs. [60, 61], as a potentially advanced feature extraction methods for speech modalities. These studies posit that PNCC and CQCC are robust to noise and channel variations with enhanced discriminative capabilities when integrated into deep learning algorithms. Therefore, these techniques could potentially enhance detection adaptability against intentional and erroneous noise in systems. It has also been noticed that current mod, such as MFCC and LFCC, generalize across languages and accents, creating further blind spots. It was also observed that while feature-based techniques like MFCC and LFCC provided foundational insights, their dependency on manual engineering limited their adaptability to novel manipulations. DL approaches using CNNs and transformers showed improved performance, but their reliance on extensive datasets introduced biases and hindered generalization. Multimodal frameworks and self-supervised models were efficient, but they were also vulnerable to adversarial attacks and noisy environments. It was also observed that there are several key indicators, like high-frequency noise, spectral discontinuities, and phonetic anomalies, that are crucial in detecting manipulations. The inconsistency in datasets, with many lacking diversity in languages, noise levels, and manipulation techniques, restricted the scalability of detection methods. Along with that, computationally intensive models, while accurate, failed to meet the demands of real-time and large-scale applications.

It is significant that AVTENet [39], although achieving perfect accuracy, is driven by multimodal approaches, integrating audiovisual features using transformers such as ViViT and AST. These methods have shown improvements in modeling cross-modal dependencies and exploiting diverse datasets. However, their reliance on well-labeled, balanced data raises concerns about real-world generalization, especially under noisy or low-resource conditions. Compared to simpler CNN-based models, their superior performance reflected enhanced feature integration but comes at the cost of increased computational complexity. Although effective, these models require more robust datasets and fine-tuned architectures to address broader manipulations and scalability challenges for practical deployment. SpecRNet, with its lightweight architecture, demonstrated scalability for real-time applications. However, it has been observed that their performance in unseen attacks and noisy real-world conditions remains underexplored. The tendency to overfit specific datasets or attack types limits broader applicability. Feature enrichment strategies have become central to advancing deepfake detection performance. Siddiqui et al. [62] propose a hybrid DenseNet121-CrossViT architecture with multi-face voting that integrates local convolutional features with global transformer attention, achieving AUC and F1 scores exceeding 0.99 across FaceForensics++, Celeb-DF, and DeepForensics 1.0, while Siddiqui et al. [63] introduce a diffusion-enhanced pipeline employing DiffPIR artifact amplification prior to DenseNet121 classification, attaining 99.91% accuracy on Celeb-DF, DFD, and proprietary datasets. While these approaches demonstrate that architectural fusion and generative preprocessing substantially improve detection accuracy under controlled benchmark conditions, both studies

Table 7 Comparative analysis of techniques algorithms and datasets mapped with performance trends

Algorithm	Techniques & models	Modality	Key features	Key indicators	Datasets	Performance	Observation	Study
CNN-Based Methods	RawNet2, SpecRNet, ResNet-18, VGG-16, Custom CNNs, MFAAN (MFCC + LFCC + Chroma)	Audio	MFCC, LFCC, Chroma, Spectrogram, LFBE	High-frequency noise, spectral discontinuities, harmonic imbalance, noise amplification, phoneme transitions	WaveFake, ASVspoof 2019, Fake-or-Real, Public Speech Sets	Accuracy: 83–98.93%, AUC up to 99.99%, EER as low as 0.04–1.26%	Strong on curated audio datasets. Performance drops under noise, codec distortion, or unseen spoof techniques. Augmentation improves robustness.	[40, 43, 50, 51, 53, 54, 58]
Transformer-Based Methods	ViT, ViViT, Dense Swin Transformer, MViT (BA-TFD+), Transformer-based patch embeddings	Audio-Visual	Mel-spectr- ograms, visual patch tokens, transformer embeddings	Spectral discontinuities, harmonic imbalance, model-specific artifacts	DFDC, FakeAV- Celeb, LAV-DF, ASVspoof (partial), VoxCeleb-based sets	AUC: 0.97–0.999, Accuracy: 92–99%, AP@0.5 = 96%	Highest performance on large multimodal datasets. Excellent global context modeling. Heavy computation and sensitive to low-quality video.	[36, 45, 48, 57]
Hybrid CNN + Transformer Approaches	Two-Plus-One Stream Network, MultimodalTrace (ResNet-1D + ResNet-3D), Hybrid ViT + CNN models	Audio-Visual	Lip-speech sync features, joint AV embeddings, CNN + Transformer fusion	Synchron- ization gaps, inconsistent dynamics, background artifacts	DFDC, FakeAV- Celeb, WLD, PDD	Accuracy: 92–97.62%, AUC up to 98%	Cross-modal synchronicity modeling improves robustness. Good cross-dataset performance. Still dependent on AV alignment quality.	[36, 37, 46]
Self-Supervised Learning (SSL)	HuBERT, XLS-R (Wav2Vec), AVForensics (SSL pretrained AV model)	Audio / Audio-Visual	SSL embeddings, frequency-aware features, large-scale pretrained audio models	Robotic cadence, limited emotional variation, phoneme transitions	AISHELL (ADD 2022), Ar-DAD Arabic, VoxCeleb2, DFDC, FF++	EER: 0.027–34% Accuracy: 94–97%, AUC: 99.9%	Excellent cross-language generalization. Stronger results when fine-tuned. Performance drops sharply on unseen noisy audio. Best scalability.	[26, 42, 57]
Multimodal (Audio-Visual) Models	AVTENet, BA-TFD+, MultimodalTrace, AVTS-DFD, CapsuleNet+Swin, AVA-CL	Audio-Visual	Mel-spectro- grams, lip-sync cues, visual transformers, contrastive embeddings	Synchron- ization gaps, temporal misalignment, emotional variation, model-specific artifacts	FakeAV- Celeb, DFDC, LAV-DF, TIMIT, FF++, WLD, PDD	Accuracy: 92–99%, AUC up to 99.9%, AP@0.5 96%	Best performance overall due to cross-modal anomaly detection. Sensitive to video compression or misalignment.	[34, 37, 38, 39, 45, 46, 47, 55, 56]
Ensemble Methods	Deepfake- Stack (Xception, ResNet101, DenseNet121, MobileNet)	Visual (used in A/V pipelines)	CNN feature maps	Spectral artifacts (via CNN models)	Face- Forensics++	Accuracy: 99.65%, AUROC: 1.0	Very high in-domain performance but weaker cross-dataset generalization. Risk of overfitting.	[35]

Table 7 (continued)

Algorithm	Techniques & models	Modality	Key features	Key indicators	Datasets	Performance	Observation	Study
ML and Feature-Based Methods	SVM, GMM, Random Forest, LFCC-GMM, TCN (baseline)	Audio	MFCC, LFCC, spectral features	High-frequency noise, spectral discontinuities, low-fidelity regions, background artifacts Limited generalization. Noise Amplification	CFAD, Fake-or-Real, WaveFake	Accuracy: 67–92%, F1: 97%	Good for baseline or simple manipulations. Performance collapses under modern generative models or noisy conditions.	[40, 44, 49, 58]

acknowledge critical limitations regarding computational expense and insufficient validation against real-world degradations such as compression artifacts, transmission noise, and variable environmental conditions. These findings further suggests that future research can prioritize computationally efficient architectures and systematic evaluation protocols that assess generalization beyond curated datasets to operational deployment scenarios.

Evaluation metrics like EER and AUC provided benchmarks, but often overlooked cross-domain generalization challenges. These findings highlighted the need for comprehensive datasets, advanced multimodal and end-to-end models, and noise-resilient frameworks to enhance detection robustness and scalability in real-world applications. Studies [42, 43] emphasized the need for hybrid approaches that integrate self-supervised learning, augmentation techniques, and multimodal analysis. Such frameworks could enhance adaptability and maintain consistent accuracy across datasets, addressing the current limitations of dataset specificity and generalization gaps. It can be stated that the prior studies often focused on algorithmic novelty or individual datasets. Their focus was too narrow, and they did not systematically evaluate the intersection of detection accuracy, dataset characteristics, and feature diversity. Our study has not only compiled results across 27 studies but also clustered and analyzed detection performance across shared datasets. The SLR has also categorized studies based on dataset type, features used, algorithmic architecture, and achieved accuracies. This provided a layered grouping strategy that provides a novelty and a cross-sectional view into how models behave differently based on dataset-specific challenges. This was an approach not prominently seen in earlier SLRs.

SLR has also focused on the feature extraction strategies in particular. While most reviews mentioned MFCC or spectrograms briefly, our SLR has classified acoustic and spectral features, learned embeddings, synchronization indicators, and cross-modal cues in technical depth. This has been done so researchers understand not just "which" features are used, but "why" some are more robust under specific conditions, such as LFCC's utility in detecting high-frequency noise artifacts versus mel-spectrograms for CNN-based classifiers. The SLR has made a strong empirical argument around the inconsistency in accuracy across datasets, even when similar techniques are applied. This is evidenced by cases such as AVFakeNet achieving 97.5% on FakeAVCeleb, but lower accuracy on ASVspoof 2019 or WaveFake datasets. This empirical observation makes a strong case that no single model maintained uniform effectiveness. This reinforced the urgent need for robust, generalizable techniques.

The SLR emphasized the necessity for hybrid models that fuse acoustic, spectral, and learned embeddings, especially when trained on augmented datasets that simulate real-world distortions such as codec compression, background noise, and adversarial inputs. Only such hybrid frameworks are likely to maintain consistency across varied scenarios.

6 Open research challenges

As highlighted in RQ6, presented in 4.6, several studies have identified some common limitations in audio deepfakes forensics. These include, but are not limited to:

- Dependency on manual feature engineering.
- Noise and Real-world condition.
- Limited generalization to unseen manipulations.

- Vulnerability to noise and real-world conditions.
- Dataset bias and insufficient diversity including cross-lingual datasets.
- Lack of robust multimodal synchronization modeling and as hybrid architectures.

Consequently, these challenges present the need for context-independent forensic solutions that are capable of identifying novel deepfakes on the fly.

One major limitation that was observed was the over-reliance on hand-crafted features such as MFCCs and LFCCs. While these features have previously supported accurate classification, they lacked robustness in modern adversarial and cross-domain conditions. Studies [35, 54] emphasized the need to shift toward adaptive, end-to-end learning systems that learn features directly from raw signals and improved flexibility against diverse attack vectors. Another limitation that has been observed was model generalization. While many frameworks performed well on the datasets they are trained on, their effectiveness became minimal on unseen manipulations or under distribution shifts [40, 49]. This highlighted the need for cross-domain learning techniques and training on diversified datasets.

Environmental noise and compression artifacts also impaired detection accuracy in practical settings. Real-world audio often includes distortions, reverberations, or low signal-to-noise ratios [43, 49]. While noise augmentation and adaptive training strategies have been proposed, these solutions are still in their early stages and require further exploration. Models trained exclusively on clean data cannot maintain accuracy under these conditions. This also gave rise to the dataset limitations. Studies [38, 41] highlighted a scarcity of high-quality, balanced, and multimodal datasets that reflect realistic manipulation scenarios. Without comprehensive and large-scale benchmarks, model performance cannot be fairly evaluated or improved systematically. Multimodal alignment inconsistencies, such as desynchronized phoneme-viseme patterns in audio-visual deepfakes, also remained difficult to detect. Methods that model fine-grained temporal dependencies across modalities [37, 39] seemed to be needed, especially as audio-visual fakes become more prevalent and convincing. The emergence of adversarial attacks has also presented challenges. Attackers can inject subtle perturbations that remain undetected by models [37]. Existing models lacked adversarial robustness, and detection systems must integrate mechanisms such as adversarial training or detection-aware data augmentation, and these areas need more research. Along with that, resource efficiency is also observed to be a significant barrier to deployment. Deepfake detection models like RawNet2 offered high accuracy their computational cost has limited their application in real-time or embedded forensic tools [53]. Research must explore lightweight architectures without compromising on performance.

The absence of standardized evaluation protocols has also impacted the progress. As the metrics for evaluation and datasets varied across studies, this has made comparison difficult. Although ADD 2022 and 2023 have proposed unified metrics such as Weighted EER and Deception Success Rate, broader adoption is still limited [37]. Therefore, a hybrid approach that combines signal processing, ML adversarial robustness, and forensic science is required to address these open challenges.

From a forensic perspective, explainable AI (XAI) will play a critical role in the future of audio deepfake detection, ensuring that system outputs are transparent, reproducible, and admissible in court. Building on approaches such as the SHAP-driven interpretability framework of Khalid et al. [64], future audio-forensic systems could highlight which acoustic cues—such as spectral irregularities, prosodic deviations, or vocoder artifacts—most influenced a deepfake classification. Likewise, the XAI-CF model proposed by Alam and Altiparmak [65] underscores the need for analyst-oriented explanations, suggesting that next-generation audio deepfake detectors should provide interpretable outputs such as attention heatmaps over manipulated segments or quantified contributions of specific frequency bands. Together, these directions indicate that the future of audio deepfake forensics lies in combining robust detection algorithms with clear, human-verifiable reasoning pathways that can withstand legal scrutiny. Furthermore, lightweight deployment is vital for supporting near real-time audio deepfake detection within digital forensic readiness, where organizations must maintain continuous, pre-incident capability to detect, preserve, and justify digital evidence. When paired with XAI, such streamlined models can provide immediate, interpretable alerts that strengthen proactive forensic workflows by ensuring that every detection event is accompanied by a transparent rationale suitable for evidentiary preservation. Ultimately, integrating efficient edge-based architectures with explainable reasoning enables systems to identify and explain manipulation attempts as they occur, enhancing both organizational readiness and the legal defensibility of collected audio evidence.

6.1 Future research opportunities

The literature identified several gaps that highlight the need for further exploration in audio deepfake detection.

Several models such as XLS-R [42], LFCC-based systems and WaveFake demonstrated lack of robustness under unseen or degraded conditions [40, 49]. This limitation indicates the need for hybrid architectures and models that can maintain stability across diverse environments. Integrating SSL with contrastive mechanisms such as temporal alignment [47] or multimodal semantic consistency [56], may create manipulation-resistant representations. Lightweight SSL variants could also address the computational challenges identified in CNN-heavy approaches such as RawNet2 and SpecRNet [40, 53].

As studies [39, 40, 49] showed declining performance on adversarial noise injection or codec distortions. Future work can explore the use of blockchain, Merkle-tree hashing, and smart-contract logging to tamper-proof audit trails in digital evidence. Watermarking within TTS pipelines for synthetic speech and embedding lightweight forensic metadata during generation could enable traceable, accountable audio outputs [17]. These techniques may be particularly relevant in scenarios where adversarial manipulation undermines purely data-driven detection systems [41, 57].

Available datasets such as DFDC, FakeAVCeleb, WaveFake, [38, 40, 49] still lack generalizability across accents, dialects, and low-resource languages. There is a need for cross-lingual detection frameworks that capture language-independent cues rather than relying heavily on language-specific spectral patterns. As datasets expand into multimodal and cross-cultural domains, there is a need for ethical and governance considerations. Studies [38] and [49] demonstrate a need to address, bias, representa-tional balance, and the responsible use of biometric speech. These gaps highlight the need for ethical frameworks that regulate how forensic datasets are collected, anno-tated, stored, and shared. Future research can ensure privacy-preserving feature extrac-tion with controlled access to biometric data, and lawful acquisition of voices, espe-cially when datasets contain public figures or re-recorded speech.

Additionally, future research can prioritize establishing forensic-oriented evaluation benchmarks through collaboration with law enforcement agencies and digital forensics practitioners to define performance metrics aligned with evidentiary standards, including false positive rates under chain-of-custody protocols, resilience against adversarial manipulation, computational efficiency for resource-constrained deployments, and explainability requirements for courtroom testimony. Furthermore, these benchmarks will be validated using a forensically curated dataset derived from verified case files with documented provenance, enabling assessment of whether audio deepfake detection systems meet the rigorous admissibility standards and operational requirements of judicial proceedings rather than laboratory conditions alone.

7 Conclusion

This study presented a comprehensive review of advancements and challenges in the field of audio deepfake detection. The study has offered critical insights into current methodologies, features, datasets, evaluation metrics, and limitations. It was observed that traditional techniques like MFCC and LFCC offered a solid foundation for distinguishing tonal and spectral characteristics, yet their manual engineering restricted adaptability to novel manipulations. DL approaches demonstrated superior accuracy on curated datasets such as FakeAVCeleb and DFDC, achieving metrics as high as 99%. However, performance inconsistencies across datasets, such as CFAD and ASVspoof 2019, highlighted the influence of dataset quality, diversity, and environmental noise on detection accuracy. Key indicators such as high-frequency noise and synchronization gaps were consistently highlighted as effective in distinguishing synthetic from authentic audio. These findings showed that existing techniques, although effective on specific datasets, lacked universal applicability across diverse audio environments. This limitation highlighted a need for innovative techniques, such as hybrid models combining feature-based and end-to-end methods, to achieve consistent high accuracy across datasets. Future work could focus on the development of generalized, noise-resilient, and scalable detection systems that are capable of achieving consistent accuracy across diverse datasets in audio deepfake detection. Also, there is a need for larger, more diverse datasets, as most existing datasets lack linguistic, acoustic, and environmental diversity. Datasets should include diverse accents, emotional tones, background noise levels, and languages to ensure broader generalization across contexts. Future work should also focus on adversarial robustness to address progressing synthetic techniques, alongside computational efficiency for scalability in real-time applications.

Author contributions

Conceptualization, R.I.; methodology, M.A., R.I.; software, M.A.; validation, R.I., M.A.; formal analysis, M.A. and R.I.; resources, M.A., R.I; data curation, M.A.; writing—original draft preparation, M.A.; writing—review and editing, R.I.; visualization, M.A., and R.I.; supervision, R.I.; project administration, R.I; All authors have read and agreed to the published version of the manuscript.

Funding

The authors received no funding for this project.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study did not require ethical approval, as no human or animal subjects were involved in the research. The authors affirm that no human research participants were involved in this study; hence, it is not applicable.

Consent for publication
The authors affirm that no human research participants were involved in this study; hence, it is not applicable.

Competing interests
The authors declare no competing interests.

Received: 31 July 2025 / Accepted: 18 March 2026
Published online: 10 April 2026

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